Graphics Programming

Uniform Data

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Uniform Variables

- Variables defined with type qualifier ***uniform***
- Used to send constant values to all vertices/fragments
  - Apply transformation to all vertices
  - Apply color to all fragments
- Uniform values can be changed between (but not during) draw calls to create animated effects and iterative programs

OpenGL Functions for Uniforms

- glGetUniformLocation(programRef, variableName)
  - Returns reference to uniform variable named *variableName* in program *programRef*
- glUniform{ 1 | 2 | 3 | 4 }{ f | i}(variableRef, value1, …)
  - Specify value of uniform variable *variableRef* in the current bound program.
    - Number (*1, 2, 3, or 4*) refers to number of values sent
    - Letter (*f or i*) refers to data type (*float* or *integer*)
      - E.g., glUniform1i, glUniform3f, etc

Graphics Programming

Clear Color Buffer

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OpenGL Functions for Clearing Windows

- Need to clear drawing area (color buffer) between renders
  - If the image contents are changing, this will avoid superimposed images
- glClearColor(red, green, blue, alpha)
  - Specify RGBA values (in range 0.0-1.0) for color used to clear color buffer
- glClear(mask)
  - Reset the contents of the buffer(s) indicated by *mask* to their default value
    - E.g., GL_COLOR_BUFFER_BIT | GL_DEPTH_BUFFER_BIT